

KYLE MALONEY

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EDUCATION

Iowa State University — Ames, IA

May 2026

Bachelor of Science in Software Engineering, Minor in Cybersecurity

GPA: 3.0

Relevant Coursework: Cloud Computing, Network Security, Cryptography, Operating Systems, Data Structures & Algorithms, Software Design

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, VHDL, SQL

Cloud & DevOps: AWS, Azure, Google Compute Engine (GCP), Kubernetes (MicroK8s), Linux, Git, Terraform

Security: Wireshark, Nmap, Terrascan, Postman

Databases: MySQL, NoSQL

PROJECTS

GridSAFE — AI-Based Anomaly Detection for Power-Grid Networks Senior Capstone | Aug 2025 — May 2026

- Built a simulated power-grid network to generate realistic traffic and anomaly data for training and evaluating a detection model.
- Developed a machine-learning anomaly-detection model using the XGBoost algorithm to flag abnormal network behavior indicative of faults or intrusions on critical infrastructure.
- Collaborated across a multi-disciplinary 5-person team through the full project lifecycle: network design, data generation, model training, and validation.

Technologies: Python, XGBoost, network simulation, machine learning

Secure Cloud File Storage — Zero-Knowledge Encrypted Sync App Academic Project | Feb 2026 — May 2026

- Designed and built a file-storage app that encrypts file contents and filenames locally before any data leaves the device, enforcing a zero-knowledge model where the server never accesses plaintext.
- Implemented secure multi-device synchronization with encrypted files hosted on AWS.
- Applied practical cryptography and cloud-security principles across the full data lifecycle.

Technologies: AWS, applied cryptography, secure sync

Self-Hosted Homelab — Media, Storage & Network Services Personal Infrastructure | Dec 2023 — Present

- Deployed and actively maintain a self-hosted homelab running a media server, network-attached storage (NAS), and AdGuard Home for network-wide DNS filtering, all orchestrated on a self-managed MicroK8s (lightweight Kubernetes) cluster.
- Administer Linux servers and manage container orchestration, networking, DNS configuration, and service uptime across the environment for users worldwide.
- Gained hands-on DevOps experience in Kubernetes-based deployment, configuration, and ongoing maintenance of self-hosted services.

Technologies: MicroK8s/Kubernetes, Linux, AdGuard Home, DNS, networking, Grafana, Prometheus

EXPERIENCE

Dining Hall Supervisor — Iowa State University Dining

Ames, IA | Oct 2022 – Feb 2026

- Supervised a team of student staff during high-volume dining service.
- Coordinated daily operations, resolved service issues in real time, and upheld health, safety, and quality standards.
- Created individualized plans for students across multiple workstations to accomplish tasks efficiently and effectively on a weekly basis.